Thesis Summary

A Novel Thermo-Mechanical Physics-Informed Neural Network for Fatigue Life Prediction in Composite Materials

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1. Motivation and Problem Statement

Fatigue life prediction in fiber-reinforced polymer (FRP) composites is a critical challenge in aerospace, automotive, and energy applications. In the very-high-cycle fatigue (VHCF) regime, conventional mechanical measurements strain histories, stiffness degradation, crack growth are often infeasible at high testing frequencies ($\sim$20 kHz). The only reliably measurable signal is surface temperature evolution, which reflects irreversible thermo-mechanical dissipation.

Existing approaches face a fundamental trade-off: classical fatigue models lack the flexibility to capture complex multi-scale composite damage, while purely data-driven ML methods overfit severely under the extreme data scarcity typical of fatigue testing (fewer than 20 experimental samples per material system). This thesis addresses both limitations through a physics-informed learning framework that embeds thermodynamic domain knowledge directly into the neural network training objective.

2. Proposed Framework

2.1 Two-Stage Modeling Pipeline

The framework follows a structured two-stage approach designed to handle heterogeneous data sources and extreme scarcity:

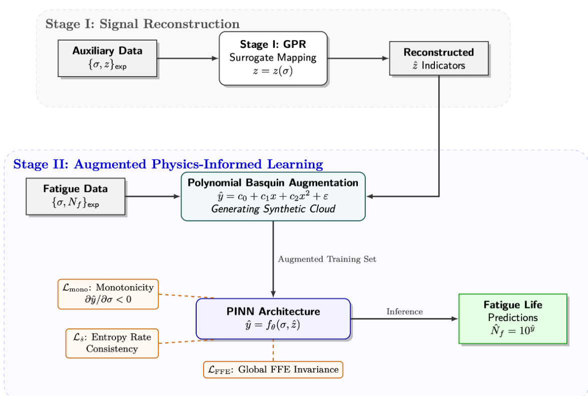

Figure 1: Your caption here

- **Stage I — Surrogate reconstruction.** Gaussian Process Regression (GPR) reconstructs a continuous, uncertainty-aware mapping from stress amplitude $\sigma_{\max}$ to stabilized thermo-mechanical dissipation indicators $z(\sigma_{\max})$ surface temperature rise, inferred plastic work, or second-harmonic thermographic signal. The fatigue-life and dissipation datasets are not reported as joint measurements. This surrogate step ensures that each fatigue-life observation is associated with a physically meaningful stabilized dissipation descriptor, even when such measurements are not directly available in the original experiments.

- **Stage II — Physics-informed fatigue regression.** A neural network $f_\theta : \mathbb{R}^2 \to \mathbb{R}$ predicts log-transformed fatigue life $\hat{y}_i = \log_{10} N_f, i$ from the input vector $\mathbf{x}_i = [\sigma_{\max}, i,\ \hat{z}(\sigma_{\max}, i)]^\top$. Data Augmentation is carried out prior to the implementation of the PINN architecture. Physical consistency is enforced through soft regularization terms added to the training objective, requiring no direct supervision of latent thermodynamic variables or full spatio-temporal temperature fields.

2.2 Physics-Informed Loss Function

The total training objective is:

$$\mathcal{L}_{\text{total}}(\theta) \;=\; \mathcal{L}_{\text{data}}(\theta) \;+\; w_B\,\mathcal{L}_{\text{Basq}}(\theta) \;+\; \lambda_1\,\mathcal{L}_{\text{mono}}(\theta) \;+\; \lambda_2\,\mathcal{L}_{\dot{S}}(\theta) \;+\; \lambda_3\,\mathcal{L}_{\text{FFE}}(\theta) \tag{1}$$

where $\mathcal{L}_{\text{data}}$ is the supervised MSE on experimental data in log-cycle space, $\mathcal{L}_{\text{Basq}}$ enforces Basquin-curve consistency on synthetically augmented samples, and $\lambda_k \geq 0$ are hyperparameters selected via nested cross-validation. The three physics terms are detailed below.

Term 1 — Monotonicity Constraints

Fatigue life is known to decrease monotonically with increasing stress amplitude and increasing dissipation indicator. Using automatic differentiation, the gradient of the predicted log-life with respect to input features is computed as $\mathbf{g}_i = \nabla_{\mathbf{x}}\,\hat{y}_i$, and positive gradients are penalized:

$$\mathcal{L}_{\text{mono}}(\theta) \;=\; \frac{1}{N}\sum_{i=1}^{N}\Big[\max\!\big(0,\ \partial\hat{y}_i/\partial\sigma_{\max}, i\big) \;+\; \max\!\big(0,\ \partial\hat{y}_i/\partial z_i\big)\Big] \tag{2}$$

This constraint prevents non-physical slope reversals and local oscillations in sparsely sampled input regions, enforcing the globally admissible geometry of the fatigue surface.

Term 2 — Entropy-Rate Consistency

Thermodynamic fatigue theory predicts that fatigue life decreases monotonically with increasing entropy generation rate $\dot{S} \propto W_p/T$, where W_p is dissipated mechanical energy per cycle and T is the stabilized specimen temperature. Differentiating the local entropy-life relationship in log-space yields the constraint:

$$\mathcal{L}_{\dot{S}}(\theta) \;=\; \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\partial\hat{y}_i}{\partial\sigma_{\max}, i} \;+\; \frac{\partial\log(W_{p,i}/T_i)}{\partial\sigma_{\max}, i}\right)^2 \tag{3}$$

This enforces local thermodynamic compatibility between predicted fatigue life and inferred dissipation trends, without requiring explicit measurement of entropy or energy variables. For the harmonic thermography dataset (Dataset 3), where signal noise is higher, this term is omitted and regularization relies on monotonicity and FFE invariance only.

Term 3 — Fracture Fatigue Entropy (FFE) Invariance

Fracture Fatigue Entropy (FFE) theory postulates that, for a given material system, fatigue failure occurs once a near-constant amount of irreversible entropy has accumulated, largely independent of stress amplitude. An accumulated dissipation proxy at failure is defined as:

$$\mathrm{FFE}_i = \frac{W_{p,i} \cdot 10^{\hat{y}_i}}{T_i} \tag{4}$$

Rather than enforcing a fixed entropy threshold (unknown a priori), FFE invariance is imposed statistically by penalizing dispersion across samples using a scale-invariant variance penalty:

$$\mathcal{L}_{\mathrm{FFE}}(\theta) = \frac{\mathrm{Var}(\mathrm{FFE}_i)}{\left(\mathbb{E}[\mathrm{FFE}_i]\right)^2 + \varepsilon} \tag{5}$$

where ε is a small numerical stabilization constant. This formulation encourages thermodynamic consistency at failure while avoiding trivial solutions such as collapse to zero life or dissipation.

2.3 Staged Training Protocol

Physics-based penalties dominate gradients if applied too early. A sequenced activation strategy resolves gradient conflicts between objectives of differing complexity:

Phase	Loss terms active	Duration	Purpose
I	$\mathcal{L}_{\mathrm{data}} + w_B \mathcal{L}_{\mathrm{Basq}}$	500 epochs	Anchor to experimental S–N trend
II	$+\lambda_1 \mathcal{L}_{\mathrm{mono}}$	500 epochs	Enforce local slope consistency
III	$+\lambda_2 \mathcal{L}_{\dot{S}}$	500 epochs	Align sensitivity with dissipation trends
IV	$+\lambda_3 \mathcal{L}_{\mathrm{FFE}}$	1500 epochs	Enforce global thermodynamic invariance

This ordering reflects increasing constraint complexity: local first-order regularization precedes global thermodynamic invariance. Ablation studies confirm that simultaneous activation from random initialization leads to unstable optimization and degraded convergence.

3. Key Results

3.1 Predictive Performance Across Three Material Systems

The framework was evaluated on three independent composite datasets covering distinct fiber architectures, matrix materials, and thermal measurement modalities. All metrics are computed in log-cycle space ($\log_{10} N_f$) under nested leave-one-out cross-validation:

Dataset	R^2	MSE	MAE	RMSE
Dataset 1 — CFRP (self-heating)	0.787	0.094	0.274	0.307
Dataset 2 — CF-PEKK (inferred dissipation)	0.912	0.066	0.218	0.257
Dataset 3 — GFRP (harmonic thermography)	0.871	0.164	0.307	0.405

All predictions remain within factor-of-two to factor-of-three error bounds across multiple decades of fatigue life. The purely data-driven baseline produced non-physical slope reversals, and instability in extreme regimes under identical training conditions.

3.2 Bayesian Variance Analysis

A Bayesian posterior analysis of residual variance confirms the statistical significance of physics-informed regularization:

- **Global analysis** (full stress domain): When analyzed globally across the full fatigue-life spectrum, the PINN demonstrates a probability of variance reduction between approximately 60–80% across the three datasets. This indicates that, in intermediate stress regimes where data coverage is relatively dense and fatigue behavior is closer to linear, the physics-based constraints provide a modest but not dominant statistical advantage over the unconstrained baseline model.
- **Extreme-regime analysis** (low-cycle + very-high-cycle fatigue): **100% posterior probability** of variance reduction across all datasets, with mean variance reduction $\Delta\sigma^2 \approx 0.156$ and ROPE (Region of Practical Equivalence) $< 1\%$. The PINN performs better significantly.

3.3 Training Dynamics

Training curve analysis reveals mechanistically interpretable behavioral transitions at each phase boundary. Activation of the FFE invariance term (Phase IV) consistently produces a discontinuous drop in out-of-sample test error alongside a temporary increase in supervised training loss reflecting a transition from local curve-fitting to globally thermodynamically admissible prediction. This behavior is reproducible across material systems and directly reflects the hierarchical constraint structure.

4. Relevance to Fraunhofer IISB Research Directions

The thesis framework was developed in the context of composite fatigue, but the underlying methodology maps directly to several of the hybrid AI–simulation directions relevant to IISB:

- **Physics-based surrogate modeling.** The two-stage GP + PINN pipeline is a concrete instance of embedding simulation-derived physical structure into a learned surrogate. The approach is directly extensible to FEM/CFD surrogate contexts where full-field simulation data is expensive and sparse sensor signals are the practical alternative.
- **Inverse problems and parameter identification.** The framework learns to infer latent material properties (dissipation behavior, accumulated entropy) from sparse indirect measurements without explicit supervision directly applicable to inverse identification of material parameters, loads, or boundary conditions from measured field data.
- **Physics-based solvers at the edge.** The trained surrogate operates at inference time with no simulation overhead, making it suitable for embedded deployment in real-time structural health monitoring and predictive maintenance architectures.
- **Digital twins.** Cross-material generalization results demonstrate that the framework recalibrates to new material systems with minimal additional data, supporting the operational update cycles characteristic of digital twin architectures.
- **Reduced-order model stabilization.** The staged constraint activation methodology enforcing physically admissible hypothesis spaces progressively is a transferable technique applicable to any domain where physics provides monotonicity, invariance, or conservation constraints as regularizers for surrogate or reduced-order models.

5. Conclusion

This thesis demonstrates that thermodynamic principles and data-driven learning are complementary tools. By embedding entropy-based invariants, monotonicity constraints, and thermodynamic consistency conditions directly into the neural network training objective, it is possible to achieve physically admissible, data-efficient, and robust fatigue-life prediction from sparse thermal measurements.

The broader contribution is a general methodology: structured, staged constraint activation as a principled approach to hybrid AI–simulation modeling, with demonstrated applicability across multiple composite material systems. The framework's design modular loss terms, uncertainty quantification, and physics-guided hypothesis space restriction is intended to generalize to surrogate modeling, digital twin development, inverse parameter identification, and real-time monitoring in industrial settings.